United States Patent Application Publication	20250282110
Kind Code	A1
Publication Date	September 11, 2025
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CARBON-FIBER-REINFORCED RESIN CYLINDER FOR PROPELLER SHAFTS

Abstract

A carbon-fiber-reinforced resin cylinder for propeller shafts includes a first-carbon-fiber-reinforced resin layer having a first carbon fiber, and a second-carbon-fiber-reinforced resin layer having a second carbon fiber and being higher in strength and lower in elastic modulus than the first-carbon-fiber-reinforced resin layer. The first carbon fiber is arranged so as to extend along a longitudinal direction of the cylinder.

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Family ID: **84487426**

Appl. No.: **18/859470**

Filed (or PCT Filed): **June 08, 2022**

PCT No.: **PCT/JP2022/023191**

Publication Classification

Int. Cl.: **B29C70/86** (20060101); **B29C70/20** (20060101); **B29K105/10** (20060101); **B29K307/04** (20060101); **B29L31/08** (20060101)

U.S. Cl.:

CPC **B29C70/86** (20130101); **B29C70/202** (20130101); B29K2105/108 (20130101); B29K2307/04 (20130101); B29K2995/005 (20130101); B29L2031/087 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a U.S. National Stage Application under 35 U.S.C § 371 of International Patent Application No. PCT/JP2022/023191 filed on 8 Jun. 2022, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to a carbon-fiber-reinforced resin cylinder for propeller shafts to be used as power transmission shafts in vehicles and the like, for example.

BACKGROUND ART

[0003] Propeller shafts typically used for automobiles are divided into two parts in the front-rear direction and include an intermediate bearing around the center in the front-rear direction. For such propeller shafts, a configuration has been known in which the two-part structure is changed to a single-part structure to improve the energy efficiency and the intermediate bearing is omitted to achieve a weight reduction. In this case, in order to shift the primary bending resonance point of the propeller shaft to a higher frequency, it is necessary to increase the diameter of its cylinder (steel tube).

[0004] Propeller shafts transmit large forces and also rotate at high speeds, and therefore require high torsional strength and bending rigidity. As for the bending rigidity in particular, increasing the diameter of a propeller shaft can increase its bending rigidity. However, increasing the diameter of a propeller shaft made of steel makes its mass large. Thus, a technique has been publicly known in which a propeller shaft is made from a carbon-fiber-reinforced resin to reduce the weight increase (see Patent Literature 1, for example).

[0005] Shaft members made of a carbon-fiber-reinforced resin are formed by shaping using a filament winding (FW) method. In this method, carbon fibers impregnated with a resin are individually wound around a core member and heated to shape the shaft member. In the case of a shaft member made of a carbon-fiber-reinforced resin, it is effective to wind carbon fibers at an angle to a rotation shaft in order to enhance the torsional strength. Also, it is publicly known to use high modulus carbon fibers in order to enhance the bending rigidity (see Patent Literature 2, for example).

CITATION LIST

Patent Literature

[0006] Patent Literature 1: JP2001-153126A [0007] Patent Literature 2: JP S61-487A

SUMMARY OF INVENTION

Technical Problem

[0008] As for the high modulus carbon fibers, there are ones made from a polyacrylonitrile material, known as PAN-based carbon fiber. However, these come with a problem of high costs. Also, it is effective to arrange carbon fibers in parallel to the rotation shaft to enhance the bending rigidity, but doing so requires arranging the carbon fibers minutely along the circumferential direction, which cannot be implemented with the filament winding method.

[0009] Incidentally, a sheet winding method has been known in which a pre-preg being a sheet-shaped intermediate material obtained by impregnating carbon fibers with a matrix resin is placed on a core. With this method, bending rigidity can be enhanced by placing and shaping the pre-preg on the core such that the carbon fibers are oriented in the direction of the rotation shaft. However, the material is costly, and the productivity is not high since layers of the sheet are laid one over another to shape the pre-preg. Thus, for products such as propeller shafts, the method is limited only to those for racing.

[0010] The present invention has been made in view of such circumstances, and an object thereof is to provide a carbon-fiber-reinforced resin cylinder for propeller shafts that can achieve high bending rigidity and high torsional strength without an increase in cost.

Solution to Problem

[0011] The present disclosure provides a carbon-fiber-reinforced resin cylinder for propeller shafts comprising: a first-carbon-fiber-reinforced resin layer having a first carbon fiber; and a second-carbon-fiber-reinforced resin layer having a second carbon fiber and being higher in strength and lower in elastic modulus than the first-carbon-fiber-reinforced resin layer, wherein the first carbon fiber is arranged so as to extend along a longitudinal direction of the cylinder.

Advantageous Effects of Invention

[0012] According to the present invention, it is possible to provide a carbon-fiber-reinforced resin cylinder for propeller shafts that can achieve high bending rigidity and high torsional strength without an increase in cost.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. **1** is a cross-sectional view schematically showing a mandrel according to a first embodiment of the present invention.

[0014] FIG. **2** is a view schematically showing a power transmission shaft manufactured using the mandrel according to the first embodiment of the present invention.

[0015] FIG. **3** is a cross-sectional view schematically showing a power transmission shaft according to the first embodiment of the present invention.

[0016] FIG. **4** is a schematic view for describing a method of manufacturing the power transmission shaft according to the first embodiment of the present invention, and is a view schematically showing a first carbon fiber layer.

[0017] FIG. **5** is a schematic view for describing the method of manufacturing the power transmission shaft according to the first embodiment of the present invention, and is a view schematically showing a second carbon fiber layer.

[0018] FIG. **6** is a schematic view for describing the method of manufacturing the power transmission shaft according to the first embodiment of the present invention, and is a view schematically showing a third carbon fiber layer.

[0019] FIG. **7** is a flowchart for describing the method of manufacturing the power transmission shaft according to the first embodiment of the present invention.

[0020] FIG. **8** is a schematic view for describing the method of manufacturing the power transmission shaft according to the first embodiment of the present invention.

[0021] FIG. **9** is a view schematically showing a first carbon fiber layer in a fiber-reinforced resin tube according to a second embodiment of the present invention.

[0022] FIG. **10** is a view schematically showing the first carbon fiber layer in the fiber-reinforced resin tube according to the second embodiment of the present invention.

[0023] FIG. **11** is a schematic view for describing a method of manufacturing a power transmission shaft according to a third embodiment of the present invention, and is a view schematically showing a first carbon fiber layer.

[0024] FIG. **12** is a schematic view for describing the method of manufacturing the power transmission shaft according to the third embodiment of the present invention, and is a view schematically showing a second carbon fiber layer.

[0025] FIG. **13** is a schematic view for describing the method of manufacturing the power transmission shaft according to the third embodiment of the present invention, and is a view schematically showing a fourth carbon fiber layer.

[0026] FIG. **14** is a schematic view for describing the method of manufacturing the power transmission shaft according to the third embodiment of the present invention, and is a view schematically showing a third carbon fiber layer.

[0027] FIG. **15** is a flowchart for describing the method of manufacturing the power transmission shaft according to the third embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0028] Embodiments of the present invention will be described in detail with reference to the drawings by taking, as an example, a case of manufacturing a power transmission shaft (propeller shaft) of a vehicle being an example a fiber-reinforced resin tube from a carbon fiber-reinforced plastic. In the following description, the same elements are denoted by the same reference signs, and overlapping description is omitted. Also, drawings to be referred to are simplified and exaggerated for clarity.

<<First Embodiment>>

[0029] As shown in FIG. **1**, a mandrel **1** according to the first embodiment is to be used to manufacture a fiber-reinforced resin tube **30**A (see FIG. **2**), and includes a mandrel body **10** and an internally fitted member **20**.

<<Mandrel Body>>

[0030] The mandrel body **10** is a resin member assuming a tubular shape. In the present embodiment, the mandrel body **10** is removed from inside the fiber-reinforced resin tube **30**A, but can remain inside the fiber-reinforced resin tube **30**A to function as a core member for the fiber-reinforced resin tube **30**A. For the mandrel body **10**, a material capable of withstanding heating for curing the resin in the fiber-reinforced resin tube **30**A can be used. Examples of such a material include PP (polypropylene resin), PET (polyethylene terephthalate resin), SMP (shape memory polymer), and so on. The mandrel body **10** integrally includes: a large diameter portion **11** at a middle portion in the axial direction; a stepped portion **12** and a small diameter portion **13** formed at an end portion on one side in the axial direction; and a tapered portion **14** and a small diameter portion **15** formed at an end portion on the other side in the axial direction. In the present embodiment, a protruding portion **16** smaller in diameter than the small diameter portion **15** is formed at the end of the small diameter portion **15** on the other side in the axial direction. The protruding portion **16** is a portion on which to fit a second metallic member **50**. Note that the mandrel body **10** may be a metallic member. Also, the mandrel body **10** as a metallic member may be configured to be pulled out of the fiber-reinforced resin tube **30**A after shaping of the fiber-reinforced resin tube **30**A.

<<Internally Fitted Member>>

[0031] The internally fitted member **20** is a tubular metallic member to be fitted into the small diameter portion **13**, which is an end portion of the mandrel body **10** on the one side in the axial direction. The internally fitted member **20** is intended to prevent radially inward deformation of the small diameter portion **13**, and a channel **20***a* for filling a pressurization fluid F (see FIG. **8**) (e.g., pressurized air) into the mandrel body **10** is formed. In the present embodiment, the pressurization fluid F is intended to pressurize the inside of the mandrel body **10** to expand it inside a shaping apparatus **100**. The pressurization fluid F is also a heating fluid for heating a thermosetting resin (a resin **34** to be described later) placed on the outer peripheral surface of the mandrel body **10** inside the shaping apparatus **100** to be described later. Note that if the mandrel body **10** is a metallic member, the internally fitted member **20** can be omitted by integrating it with the mandrel body **10**.

<Power Transmission Shaft>

[0032] As shown in FIGS. **2** and **3**, a power transmission shaft **2** manufactured using the mandrel **1** (see FIG. **1**) is a shaft that is provided in a vehicle so as to extend in its front-rear direction, and transmits power generated by a power source in the form of rotations about the axis. The power transmission shaft **2** includes the fiber-reinforced resin tube **30**A, a first metallic member **40**, the second metallic member **50**, and universal joints **3** and **4**. The fiber-reinforced resin tube **30**A is a carbon-fiber-reinforced resin cylinder for propeller shafts that transmit a propulsive force by rotating about the axis of the fiber-reinforced resin tube **30**A.

<Fiber-Reinforced Resin Tube>The fiber-reinforced resin tube **30**A is a resin-containing fiber layer formed into a tubular shape along the outer peripheral surface of the mandrel body **10**. The fiber-

reinforced resin tube **30**A is formed along the outer peripheral surfaces of the large diameter portion **11**, the tapered portion **14**, and the small diameter portion **15** of the mandrel body **10** and the outer peripheral surfaces of an end portion of the first metallic member **40** on the other side in the axial direction and an end portion of the second metallic member **50** on the one side in the axial direction. As shown in FIGS. **4** to **6**, the fiber-reinforced resin tube **30**A includes a first carbon fiber layer **31**, a second carbon fiber layer **32**, and a third carbon fiber layer **33** in this order from the inner side in the radial direction (mandrel body **10** side) as carbon fiber layers. The carbon fibers forming the second carbon fiber layer **32** and the third carbon fiber layer **33** (second carbon fibers) are higher in strength and lower in elastic modulus than the carbon fibers forming the first carbon fiber layer **31** (first carbon fibers). Incidentally, the carbon fiber layers **31**, **32**, and **33** are shown only partly in FIGS. **4** to **6**. Also, the outer peripheral surface of an end portion of the first metallic member **40** on the one side in the axial direction (an end portion located opposite from the mandrel body **10**) and the outer peripheral surface of an end portion of the second metallic member **50** on the other side in the axial direction (an end portion located opposite from the mandrel body **10**) are not covered with the fiber-reinforced resin tube **30**A and project from the fiber-reinforced resin tube **30**A.

<<First Carbon Fiber Layer>>

[0033] As shown in FIG. **4**, the first carbon fiber layer **31** is formed of a plurality of carbon fibers provided over the outer peripheral surfaces of the mandrel body **10** and the like so as to cover the mandrel body **10**. More specifically, a plurality of carbon fibers are gathered into the shape of a strip or a bundle to form a carbon fiber aggregate, and a plurality of carbon fiber aggregates are provided at different phases to form the first carbon fiber layer **31**. The carbon fibers in the first carbon fiber layer **31** are provided so as to extend in parallel to the axial direction of the mandrel body **10**. That is, the orientation angle of the carbon fibers in the first carbon fiber layer **31** with respect to an axis X of the mandrel body **10** is 0°.

[0034] The carbon fibers in the first carbon fiber layer **31** (first carbon fibers) are pitch-based carbon fibers. The first carbon fiber layer **31** is impregnated with the resin **34**, which then cures to form a single first-carbon-fiber-reinforced resin layer. The first-carbon-fiber-reinforced resin layer formed of the first carbon fiber layer **31** and the resin **34** is a straight reinforcement layer in which the first carbon fibers are arranged in parallel to the longitudinal direction of the cylinder (fiber-reinforced resin tube **30**A). The first carbon fiber layer **31** (and a fourth carbon fiber layer **35** to be described later) contributes to the eigenvalue (eigenfrequency) and bending rigidity of the fiber-reinforced resin tube **30**A.

[0035] In the first carbon fiber layer **31** (and the fourth carbon fiber layer **35** to be described later), the plurality of first carbon fibers are arranged at equal intervals in the circumferential direction. The interval between adjacent first carbon fibers can be set as appropriate. Setting a small interval between first carbon fibers adjacent to each other in the circumferential direction can increase the number of first carbon fibers in the fiber-reinforced resin tube **30**A and improve the bending rigidity of the fiber-reinforced resin tube **30**A. Also, setting a large interval between first carbon fibers adjacent to each other in the circumferential direction can decrease the number of first carbon fibers in the fiber-reinforced resin tube **30**A and reduce the cost of the fiber-reinforced resin tube **30**A.

<<Second Carbon Fiber Layer>>

[0036] As shown in FIG. **5**, the second carbon fiber layer **32** is provided on the radially outer side of the first carbon fiber layer **31** and formed of a plurality of carbon fibers provided so as to cover the first carbon fiber layer **31**. More specifically, a plurality of carbon fibers are gathered into the shape of a strip or a bundle to form a carbon fiber aggregate, and a plurality of carbon fiber aggregates are provided at different phases to form the second carbon fiber layer **32**. The carbon fibers in the second carbon fiber layer **32** are wound around the mandrel body **10** one or more times at an angle of 45° to its axial direction so as to extend helically with respect to the axial direction of the mandrel body **10**. That is, the orientation angle of the carbon fibers in the second carbon fiber layer **32** with respect to the axis X of the mandrel body **10** is 45°.

[0037] The carbon fibers in the second carbon fiber layer **32** (second carbon fibers) are polyacrylonitrile-based carbon fibers. The second carbon fiber layer **32** is impregnated with the resin **34**, which then cures to form a single second-carbon-fiber-reinforced resin layer. The second-carbon-fiber-reinforced resin layer formed of the second carbon fiber layer **32** and the resin **34** is a first bias reinforcement layer in which the second carbon fibers have an orientation angle with respect to the longitudinal direction of the cylinder (fiber-reinforced resin tube **30**A). The second carbon fiber layer **32** contributes to the torsional strength of the fiber-reinforced resin tube **30**A.

<<Third Carbon Fiber Layer>>

[0038] As shown in FIG. **6**, the third carbon fiber layer **33** is provided on the radially outer side of the second carbon fiber layer **32** and formed of a plurality of carbon fibers provided so as to cover the second carbon fiber layer **32**. More specifically, a plurality of carbon fibers are gathered into the shape of a strip or a bundle to form a carbon fiber aggregate, and a plurality of carbon fiber aggregates are provided at different phases to form the third carbon fiber layer **33**. The carbon fibers in the third carbon fiber layer **33** are wound around the mandrel body **10** one or more times at an angle of −45° to its axial direction so as to extend helically with respect to the axial direction of the mandrel body **10**. That is, the orientation angle of the carbon fibers in the third carbon fiber layer **33** with respect to the axis X of the mandrel body **10** is −45°.

[0039] The carbon fibers in the third carbon fiber layer **33** (second carbon fibers) are polyacrylonitrile-based carbon fibers. The third carbon fiber layer **33** is impregnated with the resin **34**, which then cures to form a single second-carbon-fiber-reinforced resin layer. The second-carbon-fiber-reinforced resin layer formed of the third carbon fiber layer **33** and the resin **34** is a second bias reinforcement layer in which the second carbon fibers have an orientation angle with respect to the longitudinal direction of the cylinder (fiber-reinforced resin tube **30**A) in the opposite direction to that in the first bias reinforcement layer. The third carbon fiber layer **33** contributes to the torsional strength of the fiber-reinforced resin tube **30**A.

[0040] The first carbon fibers in the first carbon fiber layer **31** (and the fourth carbon fiber layer **35** to be described later) are lower in strength and higher in elastic modulus than the second carbon fibers, and it is desirable that the tensile elastic modulus be set to 420 GPa or more and the tensile strength be set to 2600 MPa or more, for example. The second carbon fibers in the second carbon fiber layer **32** and the third carbon fiber layer **33** are higher in strength and lower in elastic modulus than the first carbon fibers and it is desirable that the tensile strength be set to 3500 to 7000 MPa and the tensile elastic modulus be set to 230 to 324 GPa, for example.

[0041] As shown in FIGS. **2** and **3**, at an end portion of the fiber-reinforced resin tube **30**A on the other side in the axial direction, a tapered portion **30***b* is formed which decreases in diameter from a large diameter portion **30***a* on the center side in the axial direction toward a small diameter portion **30***c* at the end portion on the other side in the axial direction. The large diameter portion **30***a* is a body portion assuming a shape conforming with the outer peripheral surface of the large diameter portion **11** of the mandrel body **10**. The tapered portion **30***b* assumes a shape conforming with the outer peripheral surface of the tapered portion **14** of the mandrel body **10**. The small diameter portion **30***c* is an end portion assuming a shape conforming with the outer peripheral surfaces of the small diameter portion **15** of the mandrel body **10** and part of the second metallic member **50**.

<First Metallic Member>

[0042] The first metallic member **40** is a member substantially assuming the shape of a hollow cylinder. As shown in FIG. **5** and other drawings, the first metallic member **40** is fitted to (fitted on) the mandrel body **10** at a given point during the manufacture.

[0043] The first metallic member **40** is one member of the universal joint (yoke assembly) **3** of the power transmission shaft **2**. The universal joint **3** is formed by assembling a spider, a needle bearing, and a yoke body to the first metallic member **40**.

<Second Metallic Member>

[0044] The second metallic member **50** is a member substantially assuming the shape of a solid cylinder (shaft). As shown in FIG. **5** and other drawings, the second metallic member **50** is fitted to (fitted on) the mandrel body **10** at a given point during the manufacture.

[0045] As shown in FIG. **1**, a hole portion **50***a* with a closed bottom into which the protruding portion **16** of the mandrel body **10** is insertable is formed in the end portion of the second metallic member **50** on the one side in the axial direction.

[0046] The second metallic member **50** is one member of the universal joint **4** (plunge joint assembly) of the power transmission shaft **2**. The universal joint **4** is formed by assembling a boot and a plunge joint body to the second metallic member **50**.

<Manufacturing Method>

[0047] Next, a method of manufacturing the power transmission shaft **2** using the mandrel **1** according to the first embodiment of the present invention will be described using the flowchart of FIG. **7**. The method of manufacturing the power transmission shaft **2** includes a mandrel body forming step (step S**1**), an internally fitted member placement step (step S**2**) to be executed after the mandrel body forming step, a first coupling step (step S**3**) to be executed after the internally fitted member placement step, and a second coupling step (step S**4**) to be executed after the first coupling step. The method of manufacturing the power transmission shaft **2** also includes a fiber setting step (steps S**5**A to S**5**C) to be executed after the second coupling step, and a mold loading step (step S**6**) to be executed after the fiber setting step. The method of manufacturing the power transmission shaft **2** also includes an expansion step (step S**7**) to be executed after the mold loading step, and a molding step (step S**8**) to be executed after the expansion step. The method of manufacturing the power transmission shaft **2** also includes a removal step (step S**9**) to be executed after the molding step, and a joint assembling step (step S**10**) to be executed after the removal step.

[0048] Step S**1** is a step of forming the resin mandrel body **10** shown in FIG. **1** with a shaping apparatus not shown.

[0049] After step S**1** is step S**2**, in which the internally fitted member **20** is press-fitted into the small diameter portion **13** of the mandrel body **10**. In the press-fitting, a lubricant may be applied between the outer peripheral surface of the internally fitted member **20** and the inner peripheral surface of the small diameter portion **13**. Note that step S**2** only needs to be executed before step S**8**.

[0050] After step S**2** is step S**3**, in which the first metallic member **40** is set on the end portion of the mandrel body **10** on the one side in the axial direction. In step S**3**, the first metallic member **40** is fitted to (fitted on) the stepped portion **12** of the mandrel body **10**.

[0051] After step S**3** is step S**4**, in which the second metallic member **50** is set on the end portion of the mandrel body **10** on the other side in the axial direction. In step S**4**, the second metallic member **50** is fitted to (fitted into) the protruding portion **16** of the mandrel body **10**. Here, the order of steps S**3** and S**4** can be changed as appropriate. Step S**4** may be executed first or executed at the same time.

[0052] After step S**4** is step S**5**A, in which the first carbon fiber layer **31** is formed on the outer peripheral surfaces of the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **4**. After step S**5**A is step S**5**B, in which the second carbon fiber layer **32** is formed on the outer peripheral surface of the first carbon fiber layer **31** on the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **5**. After step S**5**B is step S**5**C, in which the third carbon fiber layer **33** is formed on the outer peripheral surface of the second carbon fiber layer **32** on the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **6**. In steps S**5**A to S**5**C, the carbon fiber layers **31** to **33** are formed such that their fibers are not arranged on end portions of the first metallic member **40** and the second metallic member **50** located opposite from the mandrel **10** in the axial direction.

[0053] In steps S5A to S5C, the carbon fiber layers **31** to **33** are not fibers impregnated with a resin but are so-called raw fibers. Also, the carbon fiber layers **31** to **33** are arranged in a concurrent fashion on the outer peripheral surfaces of the end portions of the mandrel body **10**, the first metallic member **40**, and the second metallic member **50** on the other side in the axial direction by a multi-filament winding method. The carbon fiber layers **31** to **33** formed of fibers fed by the multi-filament winding method assume a so-called non-crimp structure in which they are not interwoven but formed as independent layers.
[0054] In steps S5A to S5C, the carbon fiber layers **31** to **33** are arranged on the outer peripheral surfaces of the mandrel body **10** and the like by an apparatus not shown. This apparatus is capable of setting and changing the orientation angles of the carbon fiber layers **31** to **33** as appropriate. Incidentally, the configuration may be such that the carbon fiber layers **31** to **33** are arranged on the outer peripheral surfaces of the mandrel body **10** and the like after being arranged into an integrated tubular form by the apparatus.
[0055] In a (single-)filament winding method, jigs having a plurality of radially extending pins are arranged at both end portions of a mandrel, and a single carbon fiber locked to a pin is repetitively wound around the outer peripheral surface of the mandrel to form a fiber layer. Thus, in the (single-)filament winding method, the carbon fiber layer **31**, which needs to be arranged on the mandrel body **10** and the like without being wound therearound even once, may not be held in a favorable manner on the outer peripheral surfaces of the mandrel body **10** and the like.
[0056] In contrast, in the multi-filament winding method, for each of the carbon fiber layers **31** to **33**, a plurality of carbon fibers are arranged so as to form a tubular layer and, in this state, the mandrel body **10** and the like are inserted into the tubular layer (or the tubular layer is fitted onto the mandrel body **10** and the like). Also, the multi-filament winding method can form the carbon fiber layers **31** to **33** in a concurrent fashion. Thus, with the multi-filament winding method, the carbon fiber layer **31**, which needs to be arranged on the mandrel body **10** and the like without being wound therearound even once, can be held on the outer peripheral surfaces of the mandrel body **10** and the like in a favorable manner by the carbon fiber layers **32** and **33** on the radially outer side.
[0057] After step S5C is step S**6**, in which the assembly of the mandrel **1**, the first metallic member **40**, the second metallic member **50**, and the carbon fiber layers **31** to **33** is placed inside a shaping apparatus (mold) **100**, as shown in FIG. **8**.
[0058] After step S**6** is step S**7**, in which the mandrel body **10** is expanded. As shown in FIG. **8**, in the shaping apparatus **100** in the first embodiment, a communication channel **104** is provided so as to communicate with the inside of the mandrel body **10** through the channel **20***a*. In step S**7**, the pressurization fluid F (e.g., pressurized air at 140° C. or higher) is filled into the hollow portion of the mandrel body **10** through the communication channel **104**, which is coupled to a supply apparatus not shown. The mandrel body **10** heated by the hot pressurization fluid F softens as it reaches a lower temperature (80° C., which is the transformation temperature) lower than the temperature at which the resin **34** cures, and is pressurized from inside by the pressurization fluid F to thereby undergo such an expanding deformation as to conform with the inner peripheral surface of the shaping apparatus **100**. The pressurization can prevent the mandrel body **10** from being deformed in such a direction as to become radially shrink by the resin **34** filled therearound. The pressurization can also reduce the amount of the resin **34** to be filled and thus prevent an increase in the weight of the fiber-reinforced resin tube **30**A as a finished product.
[0059] After step S**7**, the resin **34** is filled into the shaping apparatus **100**. As a result, the carbon fiber layers **31** to **33** arranged on the outer peripheral surface of the mandrel body **10** are impregnated with the resin **34**. Further, the shaping apparatus **100** is heated to cure the resin **34**, so that the fiber-reinforced resin tube **30**A is formed and also the fiber-reinforced resin tube **30**A is molded integrally with the first metallic member **40** and the second metallic member **50** (step S**8**: molding step). The resin **44** is a thermosetting resin, for example. In the present embodiment, the

mold of the shaping apparatus **100** is separated into a plurality of parts. In step S**9**, while the assembly is heated, a mold closing operation of closing the mold of the shaping apparatus **100** is also performed, and then a mold clamping operation of applying a pressure to the closed mold is performed to raise the pressure inside the mold, thereby promoting the curing of the resin **34**. Note that the mold closing operation and the mold clamping operation are performed since the present embodiment is described based on the configuration in which the mold is formed of a plurality of separated parts, but the mold clamping operation is not essential. Also, the mold closing operation and the mold clamping operation are not essential if the mold is not formed of a plurality of separated parts. Inside the shaping apparatus **100**, a space (resin pocket **102**) is formed around the outlet of a gate **101** into which the molten resin **34** is introduced. The resin **34** introduced into the shaping apparatus **100** is stored in the resin pocket **102** located on a lateral side of end portions of the carbon fiber layers **31** to **33** on the other side in the axial direction. The resin **34** stored in the resin pocket **102** is moved in the axial direction of the mandrel body **10** by vacuum suction performed from a suction port **103** formed on the opposite side from the gate **101** in the direction of arrangement of the carbon fiber layers **31** to **33** (around the outer peripheral surfaces of end portions of the carbon fiber layers **31** to **33** on the one side in the axial direction), so that the carbon fiber layers **31** to **33** are impregnated with the resin **34**. The shaping apparatus **100** is heated and also a pressure is applied to the inside of the shaping apparatus **100** in the state where the carbon fiber layers **31** to **33** are impregnated with the resin **34**. As a result, the fiber-reinforced resin tube **30**A is formed.

[0060] After step S**8** is step S**9**, in which the molded assembly, i.e., an intermediate product, is removed from the shaping apparatus **100**. After step S**9** is step S**10**, in which the universal joint **3** is attached to the intermediate product's first metallic member **40**, and the universal joint **4** is attached to the second metallic member **50**.

[0061] Incidentally, a mandrel removal step may be executed between steps S**9** and S**10**. This mandrel removal step is a step of removing the mandrel **1** out of the fiber-reinforced resin tube **30**A from the side of the first metallic member **40** with its end opening. At this time, the mandrel **1** is, for example, deformed, melted, decomposed, destructed, or eluted by a method suitable for the material used to be removed from the inside of the fiber-reinforced resin tube **30**A. This allows for a reduction in the weight of the power transmission shaft **2**.

[0062] Also, in the case of deforming the mandrel **1** to remove it from the end opening side of the first metallic member **40**, it is possible to employ, for example, a method in which the mandrel **1** is shrunk to be smaller than the end opening via depressurization of the hollow portion of the mandrel body **10** and then taken out of the fiber-reinforced resin tube **30**A.

[0063] When the mandrel removal step is performed, it is possible to depressurize the hollow portion of the mandrel body **10** through the communication channel **104** coupled to a vacuum pump not shown.

[0064] Such a mandrel removal step can be performed in a more favorable manner by plasticizing a mandrel body **10** made of a thermoplastic resin by heating or the like, for example. The mandrel removal step can also be performed in a favorable manner with a mandrel body **10** made of a thin aluminum plate with diamond patterns, for example.

[0065] The fiber-reinforced resin tube **30**A according to the first embodiment of the present invention includes the first-carbon-fiber-reinforced resin layer, which has the first carbon fibers, and the second-carbon-fiber-reinforced resin layer, which has the second carbon fibers and is higher in strength and lower in elastic modulus than the first-carbon-fiber-reinforced resin layer, and the first carbon fibers are arranged so as to extend along the longitudinal direction of the cylinder.

[0066] Thus, in the fiber-reinforced resin tube **30**A, the first carbon fibers, which have a relatively high elastic modulus, are provided so as to extend along the longitudinal direction of the cylinder. Accordingly, the second carbon fibers, which have a relatively low elastic modulus, can be the carbon fibers in the other layers. This makes it possible to ensure high bending rigidity and high torsional rigidity without causing an increase in cost.

[0067] The fiber-reinforced resin tube **30**A includes a carbon-fiber-reinforced resin layer having at least one layer of pitch-based carbon fibers as the first-carbon-fiber-reinforced resin layer, and a carbon-fiber-reinforced resin layer having at least two layers of polyacrylonitrile-based carbon fiber as the second-carbon-fiber-reinforced resin layer.

[0068] Thus, in the fiber-reinforced resin tube **30**A, the PAN-based fibers as the second carbon fibers do not need to have a high elastic modulus. This makes it possible to ensure high bending rigidity and high torsional rigidity and achieve a cost reduction in a favorable manner.

[0069] In the fiber-reinforced resin tube **30**A, the second-carbon-fiber-reinforced resin layer is provided on the radially outer side of the first-carbon-fiber-reinforced resin layer.

[0070] Thus, the second carbon fibers firmly hold the first carbon fibers from the radially outer side during the manufacture. This can improve the manufacturability of the fiber-reinforced resin tube **30**A. Also, in the fiber-reinforced resin tube **30**A, the first-carbon-fiber-reinforced resin layer is provided radially inward of the second-carbon-fiber-reinforced resin layer. This makes it possible to ensure high bending rigidity with a small cross-sectional area (less fibers), that is, at a low cost.

[0071] In the fiber-reinforced resin tube **30**A, the first-carbon-fiber-reinforced resin layer is laid by a multi-filament winding method.

[0072] Thus, during the manufacture of the fiber-reinforced resin tube **30**A, the first carbon fibers and the second carbon fibers are each such that the plurality of carbon fibers, which are to be arranged side by side in the circumferential direction, can be arranged in a single step, and the first carbon fibers can be arranged in a concurrent fashion with the second carbon fibers. This can improve the manufacturability.

[0073] Also, the second-carbon-fiber-reinforced resin layer includes the first bias reinforcement layer, in which the second carbon fibers have an orientation angle with respect to the longitudinal direction of the cylinder, and the second bias reinforcement layer, which is provided on the radially outer side of the first bias reinforcement layer and in which the second carbon fibers have an orientation angle with respect to the longitudinal direction of the cylinder in the opposite direction to that in the first bias reinforcement layer, and the first-carbon-fiber-reinforced resin layer is a straight reinforcement layer in which the first carbon fibers are arranged in parallel to the longitudinal direction of the cylinder.

[0074] Thus, it is possible to ensure high bending rigidity with the straight reinforcement layer and ensure torsional strength in two directions with the two bias reinforcement layers.

[0075] In the fiber-reinforced resin tube **30**A, the first carbon fibers are arranged at equal intervals in the circumferential direction.

[0076] Thus, the fiber-reinforced resin tube **30**A can achieve uniform bending rigidity in the circumferential direction.

<Second Embodiment>

[0077] Next, a fiber-reinforced resin tube according to a second embodiment of the present invention will be described focusing mainly on the difference from the fiber-reinforced resin tube **30**A according to the first embodiment.

[0078] As shown in FIG. **9**, in a fiber-reinforced resin tube **30**B according to the second embodiment of the present invention, the first carbon fibers in the first carbon fiber layer **31** are inclined with respect to the axial direction of the mandrel body **10** (angle θ). The inclination angle of the first carbon fibers in the first carbon fiber layer **31** is set such that a twisting torque exerted on the cylinder (fiber-reinforced resin tube **30**B) when the cylinder is subjected to the maximum rotational speed (e.g., the maximum rotational speed of the power transmission shaft **2** when a vehicle employing it moves forward) decreases the inclination angle toward a parallel angle to the longitudinal direction of the cylinder. That is, the orientation angle of the carbon fibers in the first carbon fiber layer **31** with respect to the axis X of the mandrel body **10** is preferably 0° (see FIG.

10) in a state where the cylinder is subjected to the maximum rotational speed.
[0079] In the fiber-reinforced resin tube **30**B according to the second embodiment of the present invention, the second-carbon-fiber-reinforced resin layer includes the first bias reinforcement layer, in which the second carbon fibers have an orientation angle with respect to the longitudinal direction of the cylinder, and the second bias reinforcement layer, which is provided on the radially outer side of the first bias reinforcement layer and in which the second carbon fibers have an orientation angle with respect to the longitudinal direction of the cylinder in the opposite direction to that in the first bias reinforcement layer, and the first carbon fibers in the first-carbon-fiber-reinforced resin layer are arranged so as to be inclined with respect to the longitudinal direction of the cylinder, and the inclination angle of the first-carbon-fiber-reinforced resin layer is set such that a twisting torque exerted on the cylinder when the cylinder is subjected to the maximum rotational speed decreases the inclination angle toward a parallel angle to the longitudinal direction of the cylinder.
[0080] In this way, the fiber-reinforced resin tube **30**B can achieve desired bending rigidity at the maximum rotational speed and therefore achieve improved endurance reliability.
<Third Embodiment>
[0081] Next, a power transmission shaft according to a third embodiment of the present invention will be described focusing mainly on the differences from the fiber-reinforced resin tubes **30**A and **30**B according to the first and second embodiments.
[0082] As shown in FIGS. **11** to **14**, a fiber-reinforced resin tube **30**C according to the third embodiment of the present invention further includes a fourth carbon fiber layer **35** in addition to the first carbon fiber layer **31**, the second carbon fiber layer **32**, and the third carbon fiber layer **33**.
<<Fourth Carbon Fiber Layer>>
[0083] As shown in FIGS. **11** to **14** (FIG. **13** in particular), the fourth carbon fiber layer **35** is provided on the radially outer side of the second carbon fiber layer **32** and radially inner side of the third carbon fiber layer **33**, and is formed of a plurality of carbon fibers provided so as to cover the second carbon fiber layer **32**. More specifically, a plurality of carbon fibers are gathered into the shape of a strip or a bundle to form a carbon fiber aggregate, and a plurality of carbon fiber aggregates are provided at different phases to form the fourth carbon fiber layer **35**. The carbon fibers in the fourth carbon fiber layer **35** are provided so as to extend in parallel to the axial direction of the mandrel body **10**. That is, the orientation angle of the carbon fibers in the fourth carbon fiber layer **35** with respect to the axis X of the mandrel body **10** is 0°.
[0084] The carbon fibers in the fourth carbon fiber layer **35** (first carbon fibers) are pitch-based carbon fibers. The fourth carbon fiber layer **35** is impregnated with the resin **34** (see FIG. **8**), which then cures to form a single first-carbon-fiber-reinforced resin layer. The first-carbon-fiber-reinforced resin layer formed of the fourth carbon fiber layer **35** and the resin **34** is a straight reinforcement layer in which the first carbon fibers are arranged in parallel to the longitudinal direction of the cylinder (fiber-reinforced resin tube **30**C). In the fourth carbon fiber layer **35**, the plurality of first carbon fibers are arranged at equal intervals in the circumferential direction.
[0085] The configuration may be such that the first carbon fibers in the first carbon fiber layer **31** and/or the fourth carbon fiber layer **35** in the fiber-reinforced resin tube **30**C are inclined with respect to the axial direction of the mandrel body **10**, like the first carbon fibers in the first carbon fiber layer **31** of the fiber-reinforced resin tube **30**B according to the second embodiment.
<Manufacturing Method>
[0086] Next, a method of manufacturing the power transmission shaft **2** (fiber-reinforced resin tube **30**C) using the mandrel **1** according to the third embodiment of the present invention will be described using the flowchart of FIG. **15**.
[0087] In this manufacturing method, in step S**5**A, the first carbon fiber layer **31** is formed on the outer peripheral surfaces of the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **11**. After step S**5**A is step S**5**B, in which the second carbon

fiber layer **32** is formed on the outer peripheral surface of the first carbon fiber layer **31** on the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **12**. After step S**5**B is step S**5**D, in which the fourth carbon fiber layer **35** is formed on the outer peripheral surface of the second carbon fiber layer **32** on the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **13**. After step S**5**D is step S**5**C, in which the third carbon fiber layer **33** is formed on the outer peripheral surface of the fourth carbon fiber layer **35** on the mandrel body **10**, the first metallic member **40**, and the second metallic member **50**, as shown in FIG. **14**. The fourth carbon fiber layer **35** is arranged in a non-crimp structure in a concurrent fashion with the other carbon fiber layers **31**, **32**, and **33** by a multi-filament winding method.

[0088] In this manufacturing method, in step S**8**, the first carbon fiber layer **31**, the second carbon fiber layer **32**, the fourth carbon fiber layer **35**, and the third carbon fiber layer **33** are impregnated with the resin **34** (see FIG. **8**), which is then caused to cure.

[0089] The eigenfrequency (primary bending resonance point) of the fiber-reinforced resin tube **30**C according to the third embodiment of the present invention can be set to a different value in a favorable manner without changing the thickness of the fiber-reinforced resin tube **30**C, for example, as compared to a case of simply making the first carbon fiber layer **31** thick. Also, the fiber-reinforced resin tube **30**C can achieve a desired eigenfrequency by changing the thicknesses of the first carbon fiber layer **31** and the fourth carbon fiber layer **35** as appropriate. Also, the fiber-reinforced resin tube **30**C is provided with the fourth carbon fiber layer **35** in addition to the first carbon fiber layer **31**, which makes the total thickness of so-called straight layers large. This makes it possible to ensure high bending rigidity in a more favorable manner.

[0090] While embodiments of the present invention have been described above, the present invention is not limited to the embodiments, and changes can be made as appropriate without departing from the gist of the present invention. For example, the large diameter portion (body portion) **11** of the mandrel body **10** may expand in a barrel shape which becomes smaller in diameter from a center portion of the large diameter portion **11** toward both ends or expand in a cylindrical shape with the same diameter in the axial direction. Such an expansion shape can be set as appropriate with the shape of the inner peripheral surface of the portion of the shaping apparatus (mold) **100** in which to place the large diameter portion **11**. Also, the fluid to be introduced into and filled in the mandrel body **10** may be a fluid for heating the thermosetting resin set on the outer peripheral surface of the mandrel body **10** to cure it in addition to pressurizing the inside of the mandrel body **10**. Incidentally, when the fluid is a pressurization fluid not intended for heating, the thermosetting resin will be heated by another heat source.

[0091] Also, the configuration may be such that the mandrel **1** is removed from the shaped fiber-reinforced resin tube **30**A between steps S**9** and S**10**. Also, the configuration may be such that the mandrel body **10** is removed by melting it with the heat of the resin **44** and the shaping apparatus (mold) **100** in step S**8**. The mandrel body **10** can be melted by the energies such as the heat, electricity, and vibration of other components and removed. Also, the carbon fiber layers **31** to **33** may be in a so-called crimp structure in which they are interwoven. Also, as a modification, the fibrous bodies are not limited to carbon fibers, and only need to be fibrous members capable of reinforcing resin layers (e.g., glass fibers, cellulose fibers, etc.).

Reference Signs List

[0092] **1** mandrel [0093] **10** mandrel body [0094] **30**A, **30**B, **30**C fiber-reinforced resin tube (carbon-fiber-reinforced resin cylinder for propeller shafts) [0095] **31** first carbon fiber layer (first carbon fibers, first-carbon-fiber-reinforced resin layer, straight reinforcement layer) [0096] **32** second carbon fiber layer (second carbon fibers, second-carbon-fiber-reinforced resin layer, first bias reinforcement layer) [0097] **33** third carbon fiber layer (second carbon fibers, second-carbon-fiber-reinforced resin layer, second bias reinforcement layer) [0098] **34** resin [0099] **35** fourth carbon fiber layer (first carbon fibers, first-carbon-fiber-reinforced resin layer, straight reinforcement layer)

Claims

1. A carbon-fiber-reinforced resin cylinder for propeller shafts comprising: a first-carbon-fiber-reinforced resin layer having a first carbon fiber; and a second-carbon-fiber-reinforced resin layer having a second carbon fiber and being higher in strength and lower in elastic modulus than the first-carbon-fiber-reinforced resin layer, wherein the first carbon fiber is arranged so as to extend along a longitudinal direction of the cylinder, and the second-carbon-fiber-reinforced resin layer is provided on a radially outer side of the first-carbon-fiber-reinforced resin layer

2. The carbon-fiber-reinforced resin cylinder for propeller shafts according to claim 1, comprising: a carbon-fiber-reinforced resin layer having at least one layer of a pitch-based carbon fiber as the first-carbon-fiber-reinforced resin layer; and a carbon-fiber-reinforced resin layer having at least two layers of a polyacrylonitrile-based carbon fiber as the second-carbon-fiber-reinforced resin layer.

3. (canceled)

4. The carbon-fiber-reinforced resin cylinder for propeller shafts according to claim 1, wherein the first-carbon-fiber-reinforced resin layer is laid by a multi-filament winding method.

5. The carbon-fiber-reinforced resin cylinder for propeller shafts according to claim 2, wherein the second-carbon-fiber-reinforced resin layer includes: a first bias reinforcement layer in which the second carbon fiber has an orientation angle with respect to the longitudinal direction of the cylinder; and a second bias reinforcement layer which is provided on a radially outer side of the first bias reinforcement layer and in which the second carbon fiber has an orientation angle with respect to the longitudinal direction of the cylinder in an opposite direction to the orientation angle in the first bias reinforcement layer, and the first-carbon-fiber-reinforced resin layer is a straight reinforcement layer in which the first carbon fiber is arranged in parallel to the longitudinal direction of the cylinder.

6. The carbon-fiber-reinforced resin cylinder for propeller shafts according to claim 2, wherein the second-carbon-fiber-reinforced resin layer includes: a first bias reinforcement layer in which the second carbon fiber has an orientation angle with respect to the longitudinal direction of the cylinder; and a second bias reinforcement layer which is provided on a radially outer side of the first bias reinforcement layer and in which the second carbon fiber has an orientation angle with respect to the longitudinal direction of the cylinder in an opposite direction to the orientation angle in the first bias reinforcement layer, the first carbon fiber in the first-carbon-fiber-reinforced resin layer is arranged so as to be inclined with respect to the longitudinal direction of the cylinder, and an inclination angle of the first-carbon-fiber-reinforced resin layer is set such that a twisting torque exerted on the cylinder when the cylinder is subjected to a maximum rotational speed decreases the inclination angle toward a parallel angle to the longitudinal direction of the cylinder.

7. The carbon-fiber-reinforced resin cylinder for propeller shafts according to claim 1, wherein the first carbon fiber is arranged at equal intervals in a circumferential direction.